River Test Answers

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Ferry travelling across a river(40 points)

1. (4 points) A ferry is crossing a river that is 15m wide at a velocity of $3\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $3\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

   Solution: The resultant velocity of the ferry is $4.24\,\mathrm{ms^{-1}}$ at N45.0E

   (b) The time taken to cross the river

   Solution: The time taken to cross the river is 5.0s

   (c) The displacement of the boat

   Solution: The displacement of the ferry is 21.21m at N45.0E

   (d) The vertical displacement of the boat

   Solution: The vertical displacement of the ferry is 15.0m North

2. (4 points) A ferry is crossing a river that is 192m wide at a velocity of $24\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $2\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

   Solution: The resultant velocity of the ferry is $24.08\,\mathrm{ms^{-1}}$ at N85.24E

   (b) The time taken to cross the river

   Solution: The time taken to cross the river is 8.0s

   (c) The displacement of the boat

   Solution: The displacement of the ferry is 192.67m at N85.24E

   (d) The vertical displacement of the boat

   Solution: The vertical displacement of the ferry is 16.0m North

3. (4 points) A ferry is crossing a river that is 64m wide at a velocity of $16\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $8\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is $17.89\,\mathrm{ms^{-1}}$ at N63.43E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 4.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 71.55m at N63.43E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 32.0m North

4. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $24\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $2\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is $24.08\,\mathrm{ms^{-1}}$ at N85.24E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 1.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 24.08m at N85.24E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 2.0m North

5. (4 points) A ferry is crossing a river that is 180m wide at a velocity of $18\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $7\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is $19.31\,\mathrm{ms^{-1}}$ at N68.75E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 10.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 193.13m at N68.75E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 70.0m North

6. (4 points) A ferry is crossing a river that is 100m wide at a velocity of $20\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $1\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

   Solution: The resultant velocity of the ferry is $20.02\,\mathrm{ms^{-1}}$ at N87.14E

   (b) The time taken to cross the river

   Solution: The time taken to cross the river is 5.0s

   (c) The displacement of the boat

   Solution: The displacement of the ferry is 100.12m at N87.14E

   (d) The vertical displacement of the boat

   Solution: The vertical displacement of the ferry is 5.0m North

7. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $4\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $4\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

   Solution: The resultant velocity of the ferry is $5.66\,\mathrm{ms^{-1}}$ at N45.0E

   (b) The time taken to cross the river

   Solution: The time taken to cross the river is 6.0s

   (c) The displacement of the boat

   Solution: The displacement of the ferry is 33.94m at N45.0E

   (d) The vertical displacement of the boat

   Solution: The vertical displacement of the ferry is 24.0m North

8. (4 points) A ferry is crossing a river that is 1m wide at a velocity of $1\,\mathrm{ms^{-2}}$ right relative to the water. However, the water is travelling at $1\,\mathrm{ms^{-2}}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground

   Solution: The resultant velocity of the ferry is $1.41\,\mathrm{ms^{-1}}$ at N45.0E

   (b) The time taken to cross the river

Solution: The time taken to cross the river is 1.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 1.41m at N45.0E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 1.0m North

9. (4 points) A ferry is crossing a river that is 7m wide at a velocity of $1\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $9\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is $9.06\,\mathrm{ms}^{-1}$ at N6.34E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 7.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 63.39m at N6.34E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 63.0m North

10. (4 points) A ferry is crossing a river that is 18m wide at a velocity of $18\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $4\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:

(a) The resultant velocity of the ferry relative to the ground

Solution: The resultant velocity of the ferry is $18.44\,\mathrm{ms}^{-1}$ at N77.47E

(b) The time taken to cross the river

Solution: The time taken to cross the river is 1.0s

(c) The displacement of the boat

Solution: The displacement of the ferry is 18.44m at N77.47E

(d) The vertical displacement of the boat

Solution: The vertical displacement of the ferry is 4.0m North